

1. Administrative & Study Identification

Brief Study Title: A Study of Tucatinib or Placebo With Trastuzumab and Pertuzumab for Metastatic HER2+ Breast Cancer
Full Study Title / Official Title: A Randomized, Double-blind, Phase 3 Study of Tucatinib or Placebo in Combination With Trastuzumab and Pertuzumab as Maintenance Therapy for Metastatic HER2+ Breast Cancer
Short Title/Acronym: HER2CLIMB-05
Protocol Number: SGNTUC-028
NCT Number: NCT05132582
Sponsor Name / Company: Seagen, a wholly owned subsidiary of Pfizer
Investigational Product: Tucatinib (in combination with Trastuzumab and Pertuzumab)
Phase of Development: Phase III / PHASE3
Medical Monitor: Pfizer/Seagen Clinical Operations Medical Monitor

2. Study Synopsis & Rationale

2.1 Study Objective
Primary Objective: To evaluate whether adding tucatinib to standard-of-care trastuzumab and pertuzumab improves investigator-assessed progression-free survival (PFS) as a first-line (1L) maintenance therapy in patients with HER2-positive metastatic breast cancer (MBC).
Secondary Objectives: To evaluate overall survival (OS), PFS per blinded independent central review (BICR), central nervous system (CNS) PFS, time to deterioration of health-related quality of life (QOL), pharmacokinetics (PK), and safety.

2.2 Background & Rationale To address the progression of disease that commonly occurs during standard maintenance therapy in HER2+ MBC. The current first-line (1L) standard of care (SOC) is trastuzumab plus pertuzumab and a taxane. Despite advances, most patients ultimately progress. Tucatinib, a highly selective HER2 tyrosine kinase inhibitor (TKI), previously showed significant PFS and OS benefits in heavily pretreated HER2+ MBC (HER2CLIMB). This study investigates whether introducing tucatinib earlier, as part of a 1L maintenance regimen, will delay progression while maintaining the patient's quality of life.

3. Study Design & Methodology

3.1 Design Framework
Study Type: Interventional
Control Type: Placebo-controlled
Blinding: Double-blind
Randomization: Randomized (1:1 Ratio)
Trial Status: Active, not recruiting (ACTIVE_NOT_RECRUITING)

3.2 Planned Enrollment & Duration Targeted Enrollment: 654 enrolled participants **Number of Sites:** 219 sites across 23 countries **Estimated Study Start:** March 2022 **Estimated Primary Completion:** September 2025 (Primary data cut-off)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Adults ≥ 18 years of age. 2. Centrally confirmed HER2-positive disease (unresectable locally advanced or metastatic breast cancer). 3. No evidence of progression after completing 4-8 cycles of prior 1L induction therapy (taxane + trastuzumab + pertuzumab). 4. Eastern Cooperative Oncology Group (ECOG) Performance Status of 0 or 1. 5. No brain metastases or strictly asymptomatic brain metastases (BM).

4.2 Exclusion Criteria 1. Evidence of disease progression post-1L induction therapy. 2. Prior treatment with anti-HER2 and/or anti-epidermal growth factor receptor TKIs (prior SOC for early breast cancer is permitted). 3. Inability to undergo contrast-enhanced magnetic resonance imaging (MRI) of the brain.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: * **Tucatinib/Placebo:** 300 mg taken orally twice a day (BID). * **Trastuzumab:** Intravenously (IV) at 6 mg/kg OR subcutaneously (SC) at a fixed dose of 600 mg, once every 21 days. (*Drug Details - ATC Code: L01FD01; Trade Names: Herceptin, Enhertu, Ontruzant, Trazimera, Herzuma, Phesgo, Kanjinti, Herceptin Hylecta, Ogivri, KADCYLA*) * **Pertuzumab:** IV or as a fixed-dose SC combination with trastuzumab, once every 21 days. **Route of Administration:** Oral (Tucatinib/Placebo), IV or SC (Trastuzumab/Pertuzumab). **Duration of Treatment:** Continuous in 21-day cycles until disease progression, unacceptable toxicity, or study withdrawal.

5.2 Concomitant & Prohibited Therapy Permitted: Endocrine therapy is permitted for patients with hormone receptor-positive (HR+) disease. Standard anti-emetic or anti-diarrheal supportive care. **Prohibited:** Concurrent investigational agents and other unauthorized targeted cancer therapies.

6. Study Timeline & Visit Schedule

| Visit ID | Window (Days) | Key Activities/Procedures | | :---
| :--- | :--- | | **Screening** | Day -28 to 0 | Informed Consent, Central HER2 Confirmation, Baseline Scans (incl. Brain MRI) | | **Baseline** | Day 0 | Randomization (1:1), First Dose of Tucatinib/Placebo + Antibodies | | **Interim** | Every 21 Days | Treatment administration, Safety Labs (e.g., AST/ALT monitoring), PK

sampling | | **Tumor Assessment** | Every 9-12 Weeks | Radiographic tumor assessments for PFS tracking | | **End of Study** | Variable | End of Treatment Visit (upon progression/toxicity), Safety Follow-up, Survival Follow-up |

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Progression-Free Survival (PFS) by investigator assessment per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1. Defined as the time from the date of randomization to documented disease progression or death from any cause. **Measurement Method:** RECIST v1.1 criteria using radiographic imaging.

7.2 Secondary & Exploratory Endpoints * Overall Survival (OS) (Key secondary endpoint). * PFS by Blinded Independent Central Review (BICR) per RECIST v1.1. * Central Nervous System Progression-Free Survival (CNS-PFS). * Time to deterioration of health-related quality of life (HRQoL).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine safety tracking for Treatment-Emergent Adverse Events (TEAEs). Specific protocol monitoring for known TKI toxicities including diarrhea, nausea, and elevated liver enzymes (AST/ALT). **Serious Adverse Events (SAEs):** Standard protocol for 24-hour reporting to the sponsor/regulatory agencies. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee oversees the unblinded safety and efficacy data throughout the trial's progression.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for primary efficacy analyses. Safety set for patients receiving \$\\ge 1\$ dose of study drug. **Sample Size Justification:** Target enrollment of 654 patients powered to detect a statistically significant improvement in PFS comparing the tucatinib vs. placebo arm utilizing a 2-sided stratified log-rank test. **Handling of Missing Data:** Time-to-event endpoints (PFS, OS) summarized using the Kaplan-Meier method with appropriate censoring for drop-outs or lack of progression at data cutoff.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Routine clinical monitoring including source

data verification (SDV) to ensure protocol adherence, drug accountability, and safety reporting. **Audit Trail:** eCRF entry tracking, queries, and data clarification forms (DCFs) utilized via a central Electronic Data Capture (EDC) system.

11. Study Identifiers & Governance

Posting ID: 591656080157713384 **Primary ID:** HER2CLIMB-05 (Protocol: SGNTUC-028) **Registry Number:** NCT05132582 **Sponsor/Lead Organization:** Pfizer / Seagen, a wholly owned subsidiary of Pfizer **Study Title:** HER2CLIMB-05 **Official Regulatory Title:** A Randomized, Double-blind, Phase 3 Study of Tucatinib or Placebo in Combination With Trastuzumab and Pertuzumab as Maintenance Therapy for Metastatic HER2+ Breast Cancer

12. Clinical Framework (HER2+ Breast Cancer Specific)

Disease Area: HER2 Positive Breast Cancer **Primary Condition:** Metastatic HER2-Positive Breast Cancer (MBC) **Preferred Name:** HER2-positive carcinoma of breast **UMLS Preferred Name:** Metastatic HER2-positive breast carcinoma **SNOMED ID:** 427685000 **Diagnosis Threshold:** Centrally confirmed HER2+ status (ASCO/CAP guidelines) **Medical Methodology:** 1L Maintenance Targeted Therapy **Optimization Target:** Delayed disease progression (PFS) and extended Overall Survival (OS)

13. Study Procedures & Methodology

Management Pathway: Standard clinical pathway for 1L HER2+ MBC Maintenance **Education Protocol (HCP):** Site initiation visits; protocol training for TKI dose-modifications and AE management (particularly hepatotoxicity and diarrhea) **Education Protocol (Patient):** Dosing diaries; education on concurrent oral/IV/SC regimen compliance and symptomatic self-management **Quality Audit Mechanism:** Blinded Independent Central Review (BICR) for objective radiographic efficacy auditing

14. Observation & Follow-Up Schedule

Total Duration: Continuous until disease progression or unacceptable toxicity **Onsite Visit Cadence:** Every 21 Days (Q3W) corresponding with antibody infusion/injection cycles **Remote Monitoring Frequency:** Routine site contacts; survival status tracking post-progression **Checklist Review Interval:** Radiographic tumor assessment imaging every 9 to 12 weeks

15. Sign-off & Approval

Role	Name	Signature	Date		:---	:---	:---	:---
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]					
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]					
Statistician	[Name]	_____	[DD-MMM-YYYY]					